# “Grandpa, Can You Speak Nicer?”: Envisioned Chatbot Roles and Design Tensions in Intergenerational Communication Conflicts

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## Abstract

Intergenerational conversations often break down when differences in tone, language, or expectations lead participants to feel dismissed or misunderstood. In this work, we explore how people envision AI-driven chatbot interventions for addressing communication problems in text-based intergenerational family chat. We conducted a scenario-based design interview with 10 pairs of family members from different generations, in which participants designed chatbot interventions that varied in intervention target and timing. Our findings show that participants expect chatbots to perform multiple themes of intervention, including mediating understanding, providing emotional support, offering evaluative commentary, and guiding interaction through behavioral suggestions. These expectations varied systematically across intervention contexts, giving rise to distinct chatbot roles such as neutral mediators, message coaches, repair facilitators, and emotion regulators. Across these roles, participants positioned chatbots as moral advisors that evaluate communicative appropriateness and exercise varying degrees of moral authority. Rather than prescribing specific system behaviors, this work offers a conceptual and exploratory account of AI-mediated intervention in intergenerational communication, and articulates key design tensions that arise when chatbots are imagined as socially and morally involved actors in intimate family interactions.


*DIS ’26, Singapore, Singapore*

ACM ISBN 979-8-4007-2563-0/2026/06
https://doi.org/10.1145/3800645.3813010

## CCS Concepts

• **Human-centered computing → Computer supported cooperative work**; **Empirical studies in HCI**; *Interaction design theory, concepts and paradigms.*

## Keywords

AI-mediated Communication, Intergenerational Communication, Communication Accommodation Theory, Human-AI Interaction

**ACM Reference Format:**
Tianyi Zhang, Emran Poh, Yueyue Hou, Renwen Zhang, Yi-Chieh Lee, Jiannan Li, and Anthony Tang. 2026. “Grandpa, Can You Speak Nicer?”: Envisioned Chatbot Roles and Design Tensions in Intergenerational Communication Conflicts. In *Designing Interactive Systems Conference (DIS ’26), June 13–17, 2026, Singapore, Singapore.* ACM, New York, NY, USA, 16 pages. https://doi.org/10.1145/3800645.3813010

## 1 Introduction

Communication challenges are frequently observed in intergenerational communication, which can negatively affect relationship quality and may discourage sustained engagement across age groups [63]. For instance, an older adult might use a sweeping generational generalization (e.g., “Your generation is spoiled”), or adopt an authoritative tone (e.g., “I know what’s right for you”). Conversely, a younger individual may dismiss an older family member (e.g., “You’re too old to understand this”), or use slang that an older adult would not easily comprehend (e.g., “ykyk,” “wru”). Such interactions reflect ways speakers emphasize generational differences or fail to adjust their communication to the needs and expectations of their conversational partners. In communication research, these dynamics are often described as divergence or underaccommodation [21, 22, 45, 51, 59, 63, 64].

In this work, we explore how AI-driven chatbot interventions might mitigate these types of problems in text-based intergenerational communication. While prior HCI research has largely focused on fostering bonding, enrichment, and positive experiences through technology [2, 10, 16], comparatively less attention has been paid to such everyday conversational breakdowns. These breakdowns occur in routine interactions, and over time, may discourage continued engagement between generations. Existing studies have demonstrated that chatbots can mediate certain types of communication, such as facilitating introductions between strangers [57, 58] and supporting coordination and decision making within groups [28, 35, 36]. Unlike public or organizational settings, these contexts often require intervening in emotionally charged exchanges where questions of intent, blame, and appropriateness are difficult to avoid.

However, it remains unclear how such conversational agents should be designed to address communication breakdowns in intergenerational family contexts, where relational histories, power asymmetries, and emotional stakes are deeply intertwined [12]. What positions or stances should these chatbots adopt when intervening in family conversations? For example, should a chatbot act as a neutral mediator that preserves harmony, a coach that critiques messages before they are sent, a repair facilitator that addresses harm after it occurs, or a supportive ally that helps recipients cope privately? These possibilities suggest that chatbot intervention is not a single design problem, but one of negotiating different roles that carry distinct social and moral expectations. Accordingly, we ask: How do people envision the roles chatbots should play in intergenerational communication breakdowns, and what challenges arise when these roles are enacted in different intervention contexts?

To understand this design problem, we conducted a scenario-based design study with 10 pairs of participants, each pair consisting of two family members from different generations to get cross-generational angles. We developed four conversational scenarios that depict common intergenerational communication breakdowns [30, 63, 64]. Each participant pair engaged with two of the four scenarios. To systematically explore how people envision different roles for chatbots, and how these roles vary with intervention target and timing, we asked participants to design four distinct chatbot interventions for each scenario: a publicly targeted chatbot, a pre-emptive sender-targeted chatbot that intervenes during message composition, a post-hoc sender-targeted chatbot that intervenes after message delivery, and a recipient-targeted chatbot that sends private messages to the message recipient [6, 29, 36, 57].

Our findings show that participants envision chatbots as occupying distinct interactional roles that shift across intervention contexts. These roles were realized through multiple intervention strategies, including mediating understanding, providing emotional support, offering evaluative commentary on problematic messages, and guiding interaction through behavioral guidance. Role expectations varied systematically with intervention target and timing, giving rise to patterns such as neutral mediators in public interventions, message coaches before message sending, repair facilitators after message delivery, and emotion regulators in private recipient-facing interactions. These role expectations suggest that participants expect chatbots to engage in normative evaluation, potentially positioning them as moral advisors that reason about appropriateness, responsibility, and harm within family interactions. Participants differed in the degree of moral authority they expected chatbots to exercise, ranging from gentle moral facilitation to explicit normative enforcement. Together, these findings suggest that people envision chatbots not merely as tools for communication support, but as socially and morally situated actors whose roles dynamically shift across interactional contexts within family communication.

This paper makes two contributions.

- First, we characterize how participants envision chatbots to occupy distinct interactional roles—such as mediators, coaches, repair facilitators, and emotional allies—that vary with intervention target and timing, and conceptualize how these roles are articulated through specific strategies and normative positioning.
- Second, we surface key design tensions that emerge from these role expectations, including how chatbots should make and express moral judgments, which normative values should guide chatbot behavior in family interactions, how to balance mechanical neutrality with human-like empathy, and whether interventions should rely solely on observable conversational content or incorporate bounded interpretive inference.

We emphasize that our contribution is primarily conceptual and exploratory, rather than evaluative or prescriptive.

## 2 Related Work

### 2.1 Expanding the Palette of Intergenerational Interaction

Positive intergenerational relationships have been shown to benefit both older and younger family members, supporting emotional well-being, mutual understanding, and a sense of continuity across generations [13, 25, 26]. Older and younger generations often view each other as windows into different historical and social contexts, valuing the experiences, perspectives, and knowledge exchanged through intergenerational interaction [37, 54].

Within HCI, prior work has largely focused on designing technologies that foster intergenerational bonding through shared or novel activities. These efforts include co-creative practices such as storytelling [10, 32, 61, 68], collaborative game-playing [16, 17, 19, 27, 48], and joint practices such as co-planting [47, 66]. For example, Wallbaum et al. [61] introduced StoryBox, a device that enables grandparents and grandchildren to share photos, artifacts, and audio recordings, supporting playful expression and reducing technological barriers. Derboven et al. [17] examined an intergenerational social game in which older adults and their (grand)children collaboratively engaged in a memory-based activity, strengthening social connection. More recently, Qiu et al. [47] proposed Co-Root, a collaborative planting system that allows grandparents and grandchildren to grow vegetables together at a distance, enriching communication and reinforcing emotional bonds.

While these systems successfully promote positive and engaging forms of interaction, they largely focus on creating new shared activities and pay less attention to the everyday conversational interactions through which intergenerational relationships are routinely maintained. In such everyday contexts, misunderstandings may arise inadvertently through subtle choices of wording, tone, or framing, leading to accidental tensions that can strain relationships and discourage further participation [20]. Addressing this gap, our work shifts attention from activity-centered bonding toward the communicative challenges embedded in routine family interactions, and explores how conversational agents—specifically chatbots—can be designed to mediate, mitigate, or even preempt accommodation-related tensions in everyday intergenerational communication.

## 2.2 Communication Accommodation Theory

Communication Accommodation Theory (CAT) describes how speakers adjust (or fail to adjust) their communicative behavior in response to their interlocutors with relational intent [21, 22, 59]. Non-accommodation refers to cases where speakers exaggerate insufficiently or adapt communicative styles, resulting in misunderstanding or relational strain.

Prior work has shown that non-accommodation is particularly salient in intergenerational communication, where age-related stereotypes, power asymmetries, and divergent life experiences can complicate mutual understanding [63, 64]. Two forms of non-accommodation are especially prominent in these contexts: divergence and underaccommodation.

Divergence involves communicative behaviors that accentuate perceived intergroup differences, often signaling social distance or identity assertion rather than relational alignment [20]. In intergenerational settings, older adults may enact divergence by emphasizing generational superiority or by framing younger people’s preferences and experiences as trivial or misguided, while younger speakers may diverge by foregrounding modernity in ways that implicitly exclude older adults [15, 45, 56]. Such behaviors have been linked to perceptions of impoliteness, dismissal, and reduced conversational warmth.

Underaccommodation refers to insufficient adjustment to a partner’s communicative needs or perspective [22]. In intergenerational communication, this occur by failing to acknowledge emotional significance, disregarding requests for clarification, or imposing one’s own conversational agenda [63, 64]. Underaccommodation can occur in both directions: older adults may minimize younger family members’ achievements or concerns, while younger partners may rely on slang, technical language, or culturally specific references without modification [45].

Although CAT also identifies overaccommodation as a form of non-accommodation, prior studies suggest that, in intergenerational interactions, overaccommodation is more commonly discussed in relation to speech behaviors such as exaggerated simplification, increased volume, or patronizing prosody [15]. These phenomena are less straightforward to represent in text-based exchanges and are reported less frequently in everyday digital family communication compared to divergence and underaccommodation [20]. Accordingly, our work focuses on divergence and underaccommodation as theoretically grounded and practically salient forms of intergenerational non-accommodation in mediated communication.

## 2.3 Mediating Text-based Communciation

Prior work has demonstrated the potential of chatbots to mediate text-based communication across a range of domains. Existing systems have supported interpersonal interaction by facilitating introductions between strangers [57, 58], assisted collective processes such as group decision making [9, 28, 36], and helped mitigate conflict or polarization in online discussions [24, 52, 60]. For example, IntroBot [57] was designed to help individuals familiarize themselves with one another by offering structured prompts at different stages of conversation. ArbiterBot [9] supports group decision making through real-time summarization of individual and collective viewpoints. Govers et al. [24] explored how conversational AI can mediate online debates to reduce audience polarization, applying conflict resolution strategies derived from the Thomas–Kilmann Conflict Mode Instrument (TKI) [31]. Together, these studies highlight the capacity of conversational agents to intervene meaningfully in text-based interactions by shaping how people interpret, respond to, or regulate communication.

Across this body of work, chatbot interventions vary along several key design dimensions. Some systems intervene publicly by posting messages visible to all participants in a conversation [28, 35, 36, 40]; for instance, Kim et al. [36] designed a chatbot that publicly moderates debate, with all interventions shared at the group level. Other systems adopt a more discreet approach, sending private messages to individual users [52, 57, 60]. In addition, some chatbots intervene during message composition [6, 34, 55], highlighting potentially problematic phrasing or offering revision suggestions before a message is sent, while others intervene post hoc, providing feedback only after a message has already been delivered [24, 29].

Building on this prior work, we focus on two design dimensions:

- Intervention target—whether the chatbot addresses the sender, the recipient, or the conversation as a whole—and
- Intervention timing relative to a non-accommodative message—whether the chatbot intervenes before message sending or after message delivery.

By systematically examining these dimensions, our work extends existing chatbot intervention strategies into the domain of intergenerational family communication.

# 3 Scenario-based Design Interview

Our goal was to understand how people perceive and respond to AI interventions for intergenerational communication breakdowns. In general, identifying user needs when deploying AI in complex, real-world contexts remains a challenge in the HCI community [5, 18, 41, 62, 65]. In the domain of intergenerational communication, studying accommodation problems in the lab is difficult: they are unlikely to manifest spontaneously, may be suppressed due to social desirability bias, and cannot ethically be induced through artificial stress. Traditional experimental methods do not elicit authentic experiences and attitudes toward these communication breakdowns [3, 18, 41].

Consequently, we adopted a scenario-based interview approach as a way to access experiences and reflections that are difficult to elicit through direct observation or experimentation [5]. Rather than provoking communication breakdowns, we used carefully constructed scenarios to provide a shared point of reference for discussing sensitive and situational tensions in intergenerational communication.

This approach served two complementary aims: first, we used conversational scenarios to concretize different types of accommodation breakdowns, ensuring a common understanding while prompting participants to reflect on similar moments in their own family interactions; second, we treated these scenarios as design probes, inviting participants to ideate, discuss, and critique potential chatbot interventions within specific, grounded conversational contexts.

By anchoring design activities in plausible, experience-near situations, this approach allowed participants to articulate needs and concerns, while allowing us to systematically explore reactions to a range of intervention strategies—something difficult otherwise achieve through observational or survey-based methods alone [14].

In addition, we recruited pairs of family members from different generations and conducted the interviews jointly. Intergenerational communication inherently involves at least two generational perspectives, and studying such breakdowns in isolation risks overlooking how messages are interpreted, justified, or contested across generations. By engaging family members together in the design process, we encouraged participants to consider not only their own perspectives but also how their messages and proposed chatbot interventions might be perceived by the other generation. The joint interview format further allowed us to observe how participants reacted to, negotiated, and reflected on each other's designs.

## 3.1 Chatbot Intervention Conditions

Building on prior work on chatbot intervention design [1, 8, 24, 36, 52, 55, 60], we asked participants to design chatbot interventions that varied along two dimensions: (1) *intervention target* and (2) *intervention timing* relative to a non-accommodative message.

In all scenarios, the conversation involved two family members from different generations. Within each conversation, we distinguished between the sender (who produced the non-accommodative message), and the recipient (who received it). We encode this information later in the labeling of each of the scenarios below (e.g. Scenario 1 is encoded as S1: Elder⟶Young, which means the Elder is the sender, and the Young person is the recipient).

Crossing intervention target and timing resulted in four chatbot conditions, as shown in Figure 1.

The first condition was a **publicly targeted chatbot (C1)**. After a non-accommodative message was sent, the chatbot posted a public message visible to all participants in the conversation, addressing the breakdown at the group level.

The second and third conditions were **sender-targeted chatbots**. In the **pre-emptive condition (C2)**, the chatbot intervened before message sending, providing private guidance while the sender was composing a potentially non-accommodative message. In the **post-hoc condition (C3)**, the chatbot intervened after message sending, delivering a private message to the sender once the non-accommodative message had already been posted.

The fourth condition was a **recipient-targeted chatbot (C4)**. In this case, the chatbot sent a private message to the recipient after they received the non-accommodative message.

## 3.2 Designed Scenarios: Intergenerational Non-Accommodation

To examine how intergenerational communication breakdowns might surface in everyday digital interactions, we designed four conversational scenarios depicting instances of non-accommodation between older adults and younger family members, as shown in Table 1. Each scenario was presented as a short text-based exchange between a grandparent and a grandchild. These scenarios were not intended to represent specific participant relationships, but rather to serve as design probes that foreground intergenerational communication dynamics. Participants were encouraged to interpret and relate to these scenarios based on their own experiences.

The scenarios systematically varied along two dimensions drawn from prior work on intergenerational accommodation: (1) the type of non-accommodation (divergence vs. underaccommodation), and (2) the direction of interaction (Elder⟶Young vs. Young⟶Elder). Crossing these dimensions resulted in four scenarios (S1–S4).

Specifically, the scenarios included: **S1** Elder⟶Young Divergence, **S2** Young⟶Elder Divergence, **S3** Elder⟶Young Underaccommodation, and **S4** Young⟶Elder Underaccommodation. Each scenario was grounded in familiar family contexts (e.g., food preferences, hobbies, academic achievements) to ensure plausibility and ease of participant interpretation.

## 3.3 Participants

We recruited 10 pairs of participants via social media postings in Singapore, with each pair consisting of two family members from different generations. Participants in the elder generation ranged in age from 48 to 68, while participants in the younger generation were between 18 and 35 years old. For clarity in reporting, we labeled younger participants as Y1–Y10 and elder participants as E1–E10, with matching numbers indicating participants from the same family pair. Table 2 summarizes participants' demographic characteristics.

## 3.4 Procedure

We conducted approximately 90-minute semi-structured design interviews with each participant pair. Each pair engaged with two of the four scenarios, with one scenario involving a younger sender and the other involving an elder sender. The assignment and order of scenarios for each pair are summarized in Table 2.

This design choice was informed by findings from a pilot study, which indicated that discussing a single scenario typically required approximately 40 minutes. Engaging with all four scenarios would therefore have resulted in excessively long sessions, posing challenges for participant recruitment and sustained engagement. Moreover, we sought to avoid repeatedly positioning the same generation as the message sender within a session, as this risked creating a sense of sustained blame or defensiveness among participants from that generation. To balance time constraints and relational dynamics, we therefore selected two scenarios per pair, ensuring that one involved a younger sender and the other an elder sender.

At the start of each session, we introduced the overall study procedure, and obtained informed consent, clarifying that the session would be audio-recorded. We then collected basic demographic

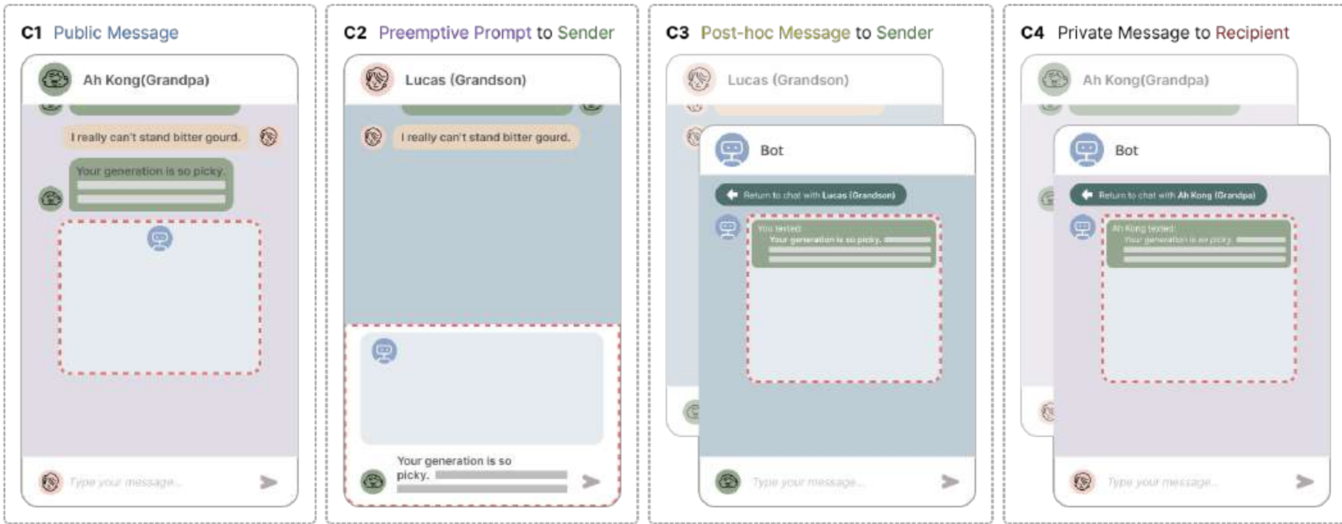


**Figure 1: The four chatbot intervention conditions used in our study, defined by intervention target (public, sender, recipient) and intervention timing (before or after message delivery). For each condition, participants were asked to design the chatbot's intervention (highlighted by the red dashed boxes), specifying what the chatbot should say or prompt in response to the non-accommodative message.**

**Table 1: Simplified versions of the four conversational scenarios used as design probes in our study. Complete scenario descriptions are available in the Appendix A.**

| | ID | Elder → Young | ID | Young → Elder |
|---|---|---|---|---|
| **Divergence** | **S1** | Ethan: *I really can't stand bitter gourd.*<br>Grandpa: *Your generation is so picky. In our time, we just ate whatever our mom cooked and didn't complain. Now you have so many choices but complain so much.* | **S2** | Grandpa: *Chloe, where did you go last night?*<br>Chloe: *I went to an underground techno show, way cooler than the dances from your era. You probably wouldn't last five minutes.* |
| **Under Accommodation** | **S3** | Amy: *Grandpa! I just found out I got the third-class scholarship this semester*<br>Grandpa: *Aim higher. You worked so hard but only got a third-class degree—how does that make sense? Why are you even happy? But if you think that's okay, then fine.* | **S4** | Grandma: *What is your thesis about?*<br>Lucas: *It's focused on optimizing a multimodal transformer. I'm improving the encoder-decoder architecture for better performance on downstream tasks.* |

and contextual information, including participants' age, upbringing, and the frequency of communication between the two family members.

For each assigned scenario, participants first read the scenario together. They were then asked to identify aspects of the interaction that might make someone feel awkward, misunderstood, or hurt. To ground discussion in lived experience, we encouraged participants to share similar situations from their own intergenerational communication.

Participants then proceeded to the design activity. The two participants were seated separately and independently designed four chatbot interventions for the scenario, corresponding to the four chatbot conditions described earlier. For each condition, participants specified what the chatbot should say or prompt in response to the non-accommodative message:

- a public message posted to the conversation after the non-accommodative message,
- a private, pre-emptive prompt shown to the sender while composing a potentially non-accommodative message,
- a private message sent to the sender after the non-accommodative message had been posted, and
- a private message sent to the recipient after receiving the non-accommodative message.

This individual design phase lasted approximately 15–20 minutes.

**Table 2: Participant demographics and communication frequency (N=20; 10 parent–child dyads). Y = younger participant; E = elder participant. Frequency categories are self-reported.**

| Elder Participants (E) | | | | Younger Participants (Y) | | | | Freq. of Text Chat | Assigned Scenarios |
|---|---|---|---|---|---|---|---|---|---|
| ID | Age | Gen. | LLM Usage | ID | Age | Gen. | LLM Usage | | |
| E1 | 59 | F | 1-2×/month | Y1 | 18 | F | 1-2×/month | Weekly | S3, S4 |
| E2 | 52 | F | <1×/month | Y2 | 21 | F | >2×/week | >2×/week | S4, S3 |
| E3 | 60 | M | <1×/month | Y3 | 27 | M | Weekly | Weekly | S3, S2 |
| E4 | 48 | F | >2×/week | Y4 | 19 | F | >2×/week | >2×/week | S2, S3 |
| E5 | 68 | F | <1×/month | Y5 | 35 | F | <1×/month | >2×/week | S3, S2 |
| E6 | 57 | F | 1-2×/month | Y6 | 26 | F | Weekly | >2×/week | S1, S4 |
| E7 | 48 | F | 1-2×/month | Y7 | 19 | F | >2×/week | >2×/week | S4, S1 |
| E8 | 59 | M | >2×/week | Y8 | 19 | F | >2×/week | >2×/week | S1, S4 |
| E9 | 60 | M | >2×/week | Y9 | 18 | F | >2×/week | >2×/week | S2, S1 |
| E10 | 63 | M | Weekly | Y10 | 35 | F | >2×/week | >2×/week | S1, S2 |

After completing their designs, the pair reconvened to present their chatbot concepts to each other, explain their design rationales, and discuss their perspectives. This joint discussion phase lasted approximately 10–15 minutes.

The same sequence of activities was then repeated for the each of the scenarios.

## 3.5 Analysis

We collected data from two primary sources: participants' designed chatbot interventions and audio recordings from discussions and interviews. In total, the dataset comprised 160 chatbot designs (40 designs for each intervention condition: public message, before message sending, after message sending, and private message to the recipient), as well as approximately 11 hours of audio-recorded discussions and interviews. All audio recordings were transcribed using Otter.ai and manually proofread to ensure accuracy.

We analyzed the data using an inductive reflexive thematic analysis approach [11]. The analysis focused primarily on participants' chatbot designs, which constituted the core analytic material, while interview transcripts were used to contextualize and interpret participants' design rationales and underlying reasoning.

The first author served as the primary analyst, conducted open coding across the dataset. To maintain analytic rigor, a second researcher reviewed the evolving codes and category definitions and provided feedback on emerging interpretations. The research team met regularly to discuss discrepancies, refine analytic boundaries, and iteratively develop the final set of themes.

# 4 Findings

## 4.1 Intervention Strategies

Before turning to the roles participants envisioned for chatbots, we first describe the intervention strategies that participants expected chatbots to perform when addressing intergenerational communication breakdowns. This interactional work included mediating how participants should understand a message, providing emotional support, offering evaluative commentary, and guiding subsequent interaction. Importantly, participants rarely treated these strategies in isolation; instead, multiple strategies were often combined within a single chatbot intervention, as shown in Figure 2.

*4.1.1 Mediating Understanding.* Participants frequently designed chatbots to help repair breakdowns by reshaping how a message should be understood, especially in situations where direct correction or confrontation might risk further conflict. Rather than evaluating whether a message was appropriate or well-intentioned, these interventions focused on reducing misunderstandings that stemmed from differences in knowledge, experience, or interpretation across generations. Participants envisioned chatbots as helping family members to re-align their understanding of each other and the messages when miscommunication might occur.

In some cases, participants treated misunderstanding as a problem of information. When one party lacked the background knowledge needed to follow the other's message, participants designed chatbots to perform what we describe as *conceptual simplification*: translating complex or unfamiliar concepts into more accessible explanations without attributing fault or intent. For example, in S4 where a younger speaker used technical terminology, Y1 designed a chatbot that explained the message to the grandmother: "Perhaps I can simplify Lucas's words for you: he is improving a large transformer AI by getting rid of unnecessary parts, which will improve its answers to visual questions." Similarly, E8 suggested that "the chatbot can explain the topic by using language that she can understand," describing the intervention as "just like translating in order for her to understand." In these designs, the chatbot functioned as a mediator that helped bridge gaps in understanding while avoiding blame.

Other cases, participants framed misunderstandings less as gaps in knowledge and more as products of generational or structural differences. Through *contextual interpretation*, chatbots were designed to explain how life experience or historical circumstances shaped the way a message was expressed or received. In S1 where a grandfather dismissed his grandson's dislike of bitter gourd as a "new world problem," Y6 designed a chatbot that explained the situation to both parties: "Grandpa grew up eating whatever was cooked, but the grandson grew up with more choices now." E9

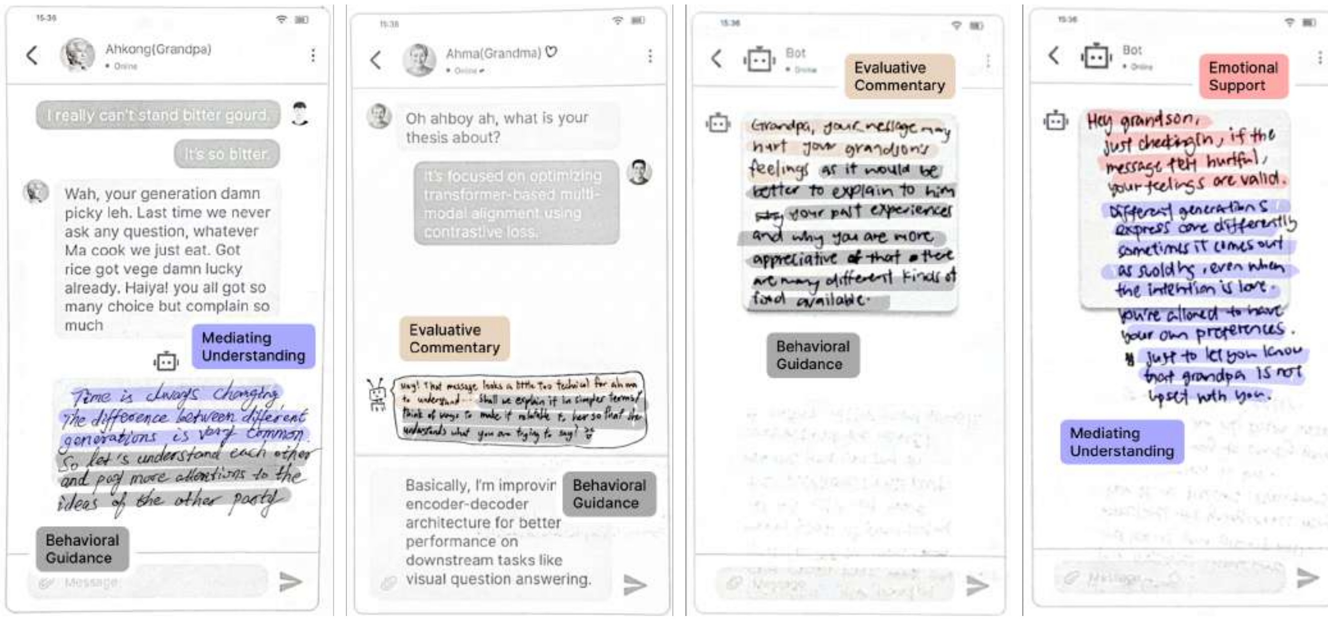


**Figure 2: Four example chatbot designs collected in the study, each corresponding to a different intervention condition. Each example demonstrates how participants combined multiple intervention strategies within a single chatbot design.**

similarly proposed that the chatbot explain that "all generations can be picky, but the older generation did not have many food choices and had limited resources." Participants perceived these background explanations as helping both sides understand each other's perspectives, thereby reducing interpersonal tension.

In emotionally charged situations, participants went a step further by designing chatbots to reshape not just what a message meant, but how it felt. Through *emotional reframing*, chatbots clarified the speaker's underlying intent in order to soften the emotional impact of a potentially hurtful message. Rather than questioning whether the message was appropriate, these interventions encouraged recipients to interpret it in a less threatening light. For example, in the S3 where a grandfather downplayed his granddaughter's scholarship achievement by urging her to "try harder," Y3 designed a chatbot that reassured the granddaughter: "Please understand that your grandpa means well and wants you to aim higher whenever possible." E2 similarly designed an intervention explaining that "he saw high potential in you and was disappointed that you didn't meet his expectations," noting that "it's helpful to know where the grandpa is coming from—even though it's hurtful, it changes how we see it." Through emotional reframing, participants positioned the chatbot as a mediator that helped recipients make sense of emotionally charged messages.

*4.1.2 Emotional Support.* Emotional support refers to chatbot interventions that address the emotional fallout of intergenerational communication breakdowns by acknowledging users' feelings and reinforcing their sense of worth or achievement. One common way participants designed chatbots to provide emotional support was through expressions of empathy, which acknowledged users' feelings. Participants designed empathetic interventions for both recipients and senders of non-accommodative messages, though with different emphases.

When empathy was directed toward the recipient, chatbots primarily validated the legitimacy of feeling hurt, awkward, or disappointed. In these cases, empathy functioned as a form of emotional validation, helping recipients feel seen and reducing the sting of a hurtful message. For example, in S2, where a granddaughter implied that her grandfather would be unable to relate to her concert experience, Y9 designed a chatbot message to the grandfather stating, "I strongly understand that your granddaughter has spoken in a terrible tone." Similarly, in S3, Y5 designed a chatbot that reassured the granddaughter: "I get how you feel after seeing this reply from Ah Kong. You may be really disappointed and hurt." These designs positioned the chatbot as an emotional witness that legitimizes recipients' feelings.

When empathy was directed toward the sender, chatbots instead acknowledged the sender's emotional motivations and framed their reactions as understandable, even if poorly expressed. Here, empathy served a different purpose: legitimizing the sender's emotional standpoint in order to reduce defensiveness and invite reflection. For instance, also in S3, E2 designed a chatbot message to the grandfather stating: "You must be happy and proud of Amy's achievement. You had high expectations for her, and when she didn't meet them, you felt disappointed." Similarly, in S1, Y6 designed a chatbot that told the grandfather, "I understand that situations back then were very difficult." In these cases, empathy served to legitimize the sender's emotional standpoint, potentially reducing defensiveness and opening space for reflection.

Words of affirmation go beyond emotional acknowledgment by explicitly reinforcing a person's value, effort, or capability. Participants expected this affirmation to restore confidence and encourage continued engagement, especially when they worried that emotional harm might lead to withdrawal from the relationship. These affirmations were usually directed toward recipients, particularly when they perceived the recipient as emotionally vulnerable. For example, in S3, E5 designed a chatbot message to the granddaughter stating: “Don’t worry, be happy. You have done your best. You are great.” In S4, Y1 designed a chatbot that affirmed the grandmother’s effort: “I think your efforts to understand Lucas better are admirable.” She explained that the goal was to encourage continued engagement: “I’m telling her that it’s great she’s asking questions and trying to understand Lucas. I’m encouraging her—like, hey, you shouldn’t let this stop you from talking to Lucas.” These designs reflect participants’ belief that emotional repair can be facilitated through explicit affirmation of personal value and effort.

In some cases, emotional support also included gentle emotion regulation. Here, chatbots encouraged users to move past distress like suggesting that they “do not take it too hard”. Rather than suppressing negative emotions, these interventions aimed to soften their impact and help users remain engaged in the relationship.

*4.1.3 Evaluative Commentary on the Message.* Participants also designed chatbots to intervene by explicitly evaluating messages, rather than merely clarifying meaning or providing emotional support. Through evaluative commentary, chatbots assessed what was problematic about a message, how it might affect others, and what consequences it could have for the relationship. These interventions marked a shift toward more overt judgment, positioning the chatbot as an analytic and normative actor within the conversation.

In some cases, participants designed chatbots to identify specific sources of tension in a message. Here, the chatbot would name the communicative feature (poor wording or poor tone) that would make the message hurtful. For example, in S2, E10 designed a chatbot message to the granddaughter stating, “Dear Chloe, we found unnecessary strong emotion in the message.” Similarly, for the same scenario, E9 designed a chatbot that told the granddaughter, “It was hurtful that you told him he was too old and not able to appreciate current music trends as you do.” Explaining this design, she noted, “I just want to tell her it’s hurtful in this sense.” By making these issues explicit, chatbots functioned as diagnostic agents that surfaced otherwise implicit sources of breakdown.

Other interventions focused less on message features and more on their likely emotional effects. By highlighting potential negative emotional, chatbots warned senders about how their messages may make recipients feel. In S3, Y4 designed a chatbot that cautioned the grandfather: “Your granddaughter may feel hurt and discouraged.” Similarly, in S2, Y5 designed a chatbot that reminded the granddaughter, “Chloe, be kind—Ah Kong might feel hurt by your message.” She described the intervention as “just a reminder to watch her choice of words.” In these cases, the chatbot acted as an affective forecaster, prompting senders to consider the emotional consequences of their communication.

Finally, some participants extended evaluation beyond immediate emotional reactions to anticipate longer-term relational consequences. For example, in S3, Y2 designed a chatbot that warned the grandfather: “It will definitely not motivate her or make her work harder. In fact, it will do the opposite!” Similarly, in S4, E1 designed a chatbot that cautioned the grandson: “Ah Ma will not be interested in further communication if you send her such a message.” Through these forward-looking evaluations, participants positioned chatbots as agents that connect present communicative choices to future relational outcomes.

*4.1.4 Behavioural Guidance.* Participants also envisioned chatbots as taking a more active role in shaping how interactions unfolded. Through behavioural guidance, chatbots moved beyond interpreting or evaluating messages to directly influencing what participants should say, do, or avoid next. These interventions positioned the chatbot as an interaction manager that could steer conversations toward repair, de-escalation, or protection.

One common form of behavioral guidance involved helping users reformulate messages. Participants designed chatbots to offer varying levels of assistance, from optional suggestions to direct rewriting of message content. For example, in S4, Y2 designed a chatbot that asked the grandson, “Would you like me to make the details of your thesis simpler so she can better understand?” In S1, Y8 designed a chatbot that advised the grandfather to “gently explain the health benefits of bitter gourd, and acknowledge that today there are many food options available.” At a higher level of intervention, chatbots directly proposed alternative message drafts. For instance, in Scenario 3, E8 designed a chatbot that suggested the grandfather send a rewritten message emphasizing both increased choice and concern for future health. Across these designs, chatbots acted as collaborators in shaping how messages were expressed.

Other interventions focused on managing the flow of conversation itself. Through interaction sequencing and repair, chatbots encouraged restraint, redirected topics, or prompted explicit repair actions such as apologies. For example, in S3, E5 designed a chatbot that advised the grandfather, “You don’t have to say anything—just listen to her sharing with you.” Other designs suggested redirecting the conversation to a less contentious topic. In S1, E7 designed a public chatbot message proposing a shift in focus: “There are many food options today that can provide the same nutritional needs. Would you like me to list some options?” In contrast, repair-oriented interventions addressed existing harm by explicitly prompting corrective actions. For instance, in S1, E6 designed a chatbot that instructed the grandfather to apologize: “Please apologize by saying: ‘I am sorry. I shouldn’t have said such words.’” Here, chatbots guided not just what was said, but when and how interaction should proceed.

Finally, some participants designed chatbots to help users set boundaries when continued engagement risked further harm. These boundary-setting interventions encouraged disengagement, delay, or explicit expression of emotional limits. One approach involved encouraging disengagement or delay. For example, in S2, E3 designed a chatbot that advised the grandfather: “Avoid participation, Grandpa. It’s hurting your feelings.” Another approach supported recipients in asserting their own boundaries by expressing their feelings. In S3, Y5 designed a chatbot that offered to help the granddaughter articulate her emotions to her grandfather, stating, “Let me tell you how to express your feelings to Grandpa.” Rather than resolving the communicative issue directly, these interventions prioritized emotional protection and well-being.

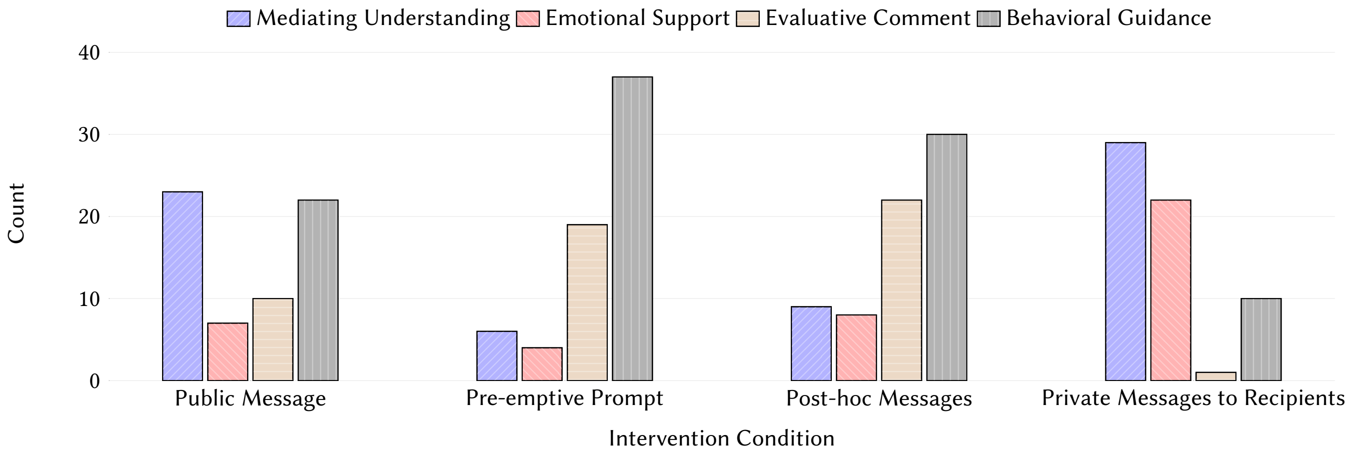


**Figure 3: Distribution of message categories across Intervention Condition**

## 4.2 How Intervention Varies by Chatbot Intervention Condition

To examine how participants' expectations for chatbots varied across intervention conditions, we analyzed how the intervention strategies identified in Section 4.1 clustered across the predefined chatbot conditions. Although these conditions were fixed by the study design, participants articulated systematically different expectations for what chatbots should do in each case. When particular strategies repeatedly co-occurred within a condition, participants treated the chatbot as playing a distinct interactional role. As shown in Figure 1, these role expectations were calibrated to where and how the chatbot intervened in the conversation.

*4.2.1* ***Neutral Mediators*** *in Public Message Interventions.* In public, shared conversational spaces, participants expected chatbots to act as neutral mediators whose primary responsibility was to preserve face and maintain conversational harmony. In these public interventions, chatbots were primarily designed to mediate understanding and provide behavioral guidance, while emotional support and evaluation were comparatively rare.

In this context, mediating understanding mainly took the form of emotional reframing and contextual interpretation aimed at helping both parties recognize each other's perspectives. Rather than taking sides or evaluating blame, participants expected public-facing chatbots to frame disagreements in ways that allowed both parties to maintain dignity. For example, in S3, Y5 designed a chatbot that posted the following public message: "Ah Kong, I sense your disappointment here because your granddaughter failed to meet your expectations. However, she managed to get a highly competitive scholarship—that's already an achievement worthy of praise, correct?" Similarly, in S1, E8 designed a public chatbot message stating: "Time is always changing. Differences between generations are common, so let's try to understand each other." Explaining this design, the participant noted that encouraging mutual understanding "will reduce miscommunication between generations and help move the conversation toward points both sides can agree on."

In public interventions, behavioral guidance was predominantly oriented toward maintaining conversational harmony rather than correcting individual behavior. Participants designed chatbots to redirect topics, regulate conversational flow, or otherwise de-escalate tension. For instance, in S1, E6 designed a chatbot that asked publicly, "So what kind of vegetables do you like to eat? Next time we can cook that." She explained that "the chatbot can say something that both Ah Kong and the grandson would like to hear." In contrast, for S2, Y4 designed a chatbot message stating, "This discussion may have become heated. Would you like to continue the chat?" She reasoned that "the conversation was getting a bit offensive, so the chatbot should prompt them to stop the conversation here first."

Taken together, these designs show that in shared conversational spaces, participants expected chatbots to prioritize moderation and harmony preservation. Accordingly, public interventions focused on de-escalating tension and preserving face, rather than offering emotional support or making explicit moral judgments about either party.

*4.2.2* ***Message Coaches*** *in Private Pre-emptive Prompts to Sender.* Before message sending, participants envisioned chatbots as message coaches that help senders anticipate problems and adjust their wording before harm occurs, rather than managing conflict after it emerges. Interventions in this condition emphasized a combination of evaluative commentary and behavioral guidance, which were often tightly coupled within a single interaction.

In these pre-emptive interventions, chatbots typically began by diagnosing why a draft message might be non-accommodative. For example, in S4, E6 designed a chatbot that alerted the sender, "That is too difficult for laymen to understand," while the message was still being composed. The chatbot then followed up by suggesting concrete revision strategies, advising the sender to "rephrase it to make it simpler, or give examples or pictures to illustrate." Similarly, in S1, Y9 designed a chatbot that commented on the sender's tone before message delivery: "Your tone seems a bit harsh. Would you like to make the conversation a bit more light-hearted and caring?"

Across these designs, participants framed pre-sending chatbot interventions as preventative, with chatbots diagnosing issues early and offering actionable guidance to help senders avoid introducing problematic content that could lead to emotional harm or relational strain.

#### 4.2.3 **Repair Facilitators** *in Private Post-hoc Messages to Sender.*

After a message had been sent, participants' expectations shifted toward repair facilitators that help diagnose harm and guide relational recovery. Although evaluative commentary and behavioral guidance remained the most common intervention themes in this condition, they were reoriented toward addressing harm that had already occurred.

In post-sending interventions, chatbots were expected to first diagnose the potential emotional or relational damage caused by the message, and then suggest concrete reparative actions. These actions often included issuing apologies, providing clarifications, or reframing the original intent. For example, in S1, after the sender had posted a non-accommodative message, Y4 designed a chatbot that first warned the sender, "Your message may have offended your grandpa's feelings," and then prompted reparative steps by suggesting either a softer rewritten version of the message or a follow-up apology. Similarly, in the same scenario, E8 designed a chatbot that told the sender, "Your message may hurt your grandson's feelings," and encouraged the sender to explain past experiences and articulate why they were more appreciative of the food options available today.

Across these designs, participants consistently framed the chatbot as a repair facilitator that supports relational recovery rather than message optimization. By explicitly naming the harm and guiding senders toward reparative responses, chatbots were expected to help restore relational alignment after a breakdown had already taken place.

#### 4.2.4 **Emotion Regulators** *in Private Messages to Recipients.*

In private, recipient-facing interventions, participants framed chatbots as emotion regulators that help recipients cope with and recover from hurtful, dismissive, or discouraging messages. To achieve this goal, participants frequently combined emotional support or mediating understanding.

Emotional support served as the most direct mechanism for alleviating distress. For example, in S3, E2 designed a chatbot that sent the recipient a private message stating: "Congrats, Amy. Proud of you. Your hard work and continuous effort paid off. It's very hard to hear what Grandpa said." Explaining this design, the participant noted that "she's already heard the negative message, so hearing appreciation from anyone can change the mood." Such interventions positioned the chatbot as an affective ally that acknowledges emotional pain while affirming the recipient's value.

In addition to direct emotional support, participants often designed chatbots to mediate understanding by clarifying the sender's underlying intent, thereby reframing the message as less personally harmful than it initially appeared. For instance, in S2, Y3 designed a chatbot that explained to the recipient: "The granddaughter will not want you to come down and not enjoy yourself." The participant explained that the purpose of this intervention was "to let Grandpa know the granddaughter's good intention, which is that she didn't want to waste his time." By contextualizing the sender's intent, these interventions aimed to reduce the emotional intensity of the recipient's interpretation.

Here, private, recipient-facing interventions prioritized emotional relief, validation, and coping within a safer, non-public space. Together, these designs suggest that participants treated emotional regulation as being important for continued interaction, especially in relationships where disengagement was not seen as a realistic option.

## 4.3 Chatbots as Moral Advisors

What we see, then, is that participants consistently expected chatbots to do more than manage conversational flow or reduce misunderstanding. Instead, chatbots were often positioned to evaluate whether messages were appropriate, who bore responsibility for harm, and how participants ought to respond. In doing so, participants treated chatbots not as neutral tools, but as morally situated actors capable of making and expressing moral judgments. We describe this positioning as chatbots acting as *moral advisors*: agents expected to reason about intent and impact, take sides when necessary, and intervene normatively in communication breakdowns.

In this subsection, we unpack how participants articulated this moral positioning along two dimensions: (1) moral alignment, or whose side the chatbot takes when evaluating a communicative breakdown, and (2) degree of moral authority, or how strongly the chatbot is expected to assert normative judgments. Together, these analyses reveal how participants envision chatbots not only as interactional mediators, but as morally situated actors whose authority and alignment are carefully calibrated within family communication contexts.

#### 4.3.1 *Moral Alignment: Whose Side the Chatbot Takes.*

Moral alignment concerns the outcome of the chatbot's moral evaluation. More specifically, whose perspective the chatbot ultimately supports. It articulates implicit or explicit judgments about responsibility, intent, and harm. Participants' designs varied in whether the chatbot aligned primarily with the sender, the recipient, or attempted to balance both perspectives.

**Sender-Oriented Moral Alignment.** In sender-oriented moral alignment, chatbots were designed to align normatively with the sender, often appearing on the surface to "speak on behalf of" the sender. However, rather than simply defending the sender, these interventions performed a form of moral justification by reframing a potentially hurtful message as morally understandable given the sender's intentions or circumstances. For example, in S3, E1 designed a chatbot message to the recipient stating: "Don't be disheartened. Your grandpa is placing high hopes on you. He expects better achievements because he believes you can do it." Similarly, in S1, E7 designed a chatbot that explained the grandfather's message to the recipient by emphasizing its underlying motivation: "Bitter gourd is nutritional! Grandpa wants you to be healthy and grow resilient in all circumstances." Taken together, these designs suggest that participants treated non-accommodative messages from family members as morally interpretable through a lens of care rather than hostility. Even when a message was experienced as hurtful, participants assumed that its origin within a familial relationship made it more likely to be motivated by concern, expectation, or responsibility rather than malicious intent. Within this moral framing, participants implicitly prioritized intention over consequence: if the sender's motivation could be understood as reasonable or well-intentioned, the moral blame associated with the negative emotional impact was seen as mitigable rather than definitive.

**Recipient-Oriented Moral Alignment.** In recipient-oriented moral alignment, chatbots were designed to align normatively with the recipient by explicitly acknowledging harm and validating the recipient's emotional experience. Unlike sender-oriented alignment, which often emphasizes intention, these interventions foregrounded the moral impact of the message and treated the recipient's feelings as morally legitimate. When aligning with recipients, chatbots frequently articulated clear moral judgments that framed the sender's behavior as inappropriate or harmful. For example, in S2, Y9 designed a chatbot that told the sender: "I believe that your message is quite harsh and painful toward your grandfather." This intervention explicitly evaluated the message as morally problematic due to the harm it could cause. In other cases, recipient-oriented alignment focused on affirming the moral legitimacy of the recipient's emotional response. For instance, in S1, Y6 designed a chatbot message to the recipient stating: "If the message felt hurtful, your feelings are valid, because different generations express care differently." Here, the chatbot simultaneously identified the moral impact of the message and validated the recipient's reaction as reasonable. Across these examples, recipient-oriented moral alignment positioned the chatbot as a moral advocate for those who were harmed in the interaction.

**Dual Moral Alignment.** In dual moral alignment, chatbots were designed to acknowledge and balance the moral claims of both sender and recipient. Rather than offering unilateral moral justification or condemnation, these interventions engaged in explicit moral reasoning by recognizing the legitimacy of competing perspectives. Typically, chatbots first validated both parties' positions before articulating constraints on acceptable behavior. For example, in Scenario 3, Y1 designed a chatbot that stated: "While I understand that you believe she should have achieved more, you should not project your expectations onto her." Similarly, in the same scenario, E2 designed a chatbot message acknowledging both sides: "I understand, Grandpa, that you had high expectations for Amy. Third class is also her achievement." In these designs, the chatbot did not avoid moral judgment; instead, it made that judgment explicit by weighing competing moral considerations, such as intention versus impact, expectations versus autonomy, and aspiration versus emotional harm.

*4.3.2 Degree of Moral Authority.* Because moral evaluation inherently involves judgments of right and wrong, participants' designs of moral alignment were closely intertwined with their expectations of how much moral authority chatbots should exercise. Across scenarios, participants varied in whether they envisioned chatbots as gentle facilitators of moral reflection or as authoritative agents that explicitly enforce normative standards.

Some participants designed chatbots to exercise low moral authority, relying on hedging, suggestions, and possibility-oriented language to prompt reflection without issuing direct condemnation. For example, in Scenario 4, E6 designed a chatbot that gently alerted the sender, "This statement may be difficult for her to understand." In the same scenario, Y2 proposed an even softer formulation framed as an offer of assistance: "Would you like me to make the details of your thesis simpler so she can better understand?" Similarly, in Scenario 1, E8 designed a chatbot that highlighted the potential benefit of revising the message, noting that "in this way, he may be more willing to accept your advice." These interventions positioned the chatbot as a reflective guide that preserves the sender's agency while encouraging moral consideration.

In contrast, other participants expected chatbots to exercise high moral authority by explicitly labeling behaviors as inappropriate, unfair, or rude, often coupled with strong behavioral directives. For instance, in Scenario 1, Y7 designed a chatbot that directly confronted the sender: "Hi, Ah Kong, you are very rude leh. Last time when you were at Ethan's age, if Ma talked to you like that, would you like it?" Similarly, E6 designed a chatbot that unambiguously instructed the sender to apologize: "Please apologize by saying: 'I am sorry. I shouldn't have said such words.'" In these designs, the chatbot functioned as a moral authority that not only evaluated wrongdoing but also prescribed corrective action.

## 5 Discussion

Our findings highlight that designing chatbot interventions for intergenerational communication is not simply a matter of improving message clarity or reducing misunderstandings. Instead, participants' designs reveal deeper questions about the social and moral roles chatbots are expected to play when intervening in emotionally charged, relational contexts. In this section, we discuss the broader implications of these findings, focusing on the normative values chatbots may encode, the risks and benefits of empathetic and inferential interventions, and the limits of AI mediation in supporting intergenerational relationships.

### 5.1 How Should Chatbots Make Moral Judgments?

Across our study, participants frequently positioned chatbots to evaluate the moral dimensions of intergenerational communication breakdowns—assessing whether a message was appropriate, who bore responsibility for harm, and how participants ought to respond. In other words, participants often expected chatbots to engage in moral judgment when mediating intergenerational conflict. This observation aligns with prior work suggesting that AI systems have the potential to function as "artificial moral experts," given their capacity for logical reasoning, access to supporting information, and ability to generate actionable guidance [49]. Relatedly, Giubilini and Savulescu [23] argue that because AI systems are not distorted by emotional bias, they may, in some contexts, improve the quality of moral decision making without undermining human agency.

At the same time, delegating moral judgment to chatbots introduces significant challenges. Normative evaluation is inherently subjective [38], and the specific moral norms encoded in a system can profoundly shape user experience. As we discuss in Section 5.2, these normative choices are especially consequential in intergenerational family communication, where expectations about respect, care, and authority vary widely across individuals and cultures. Moreover, prior research has shown that algorithmic decisions are often perceived as less fair, less trustworthy, and more emotionally alienating than human judgments [39, 43]. Such concerns may be amplified in intergenerational contexts, where communication involves subtle emotional cues, relational histories, and deeply personal values. In such contexts, AI interventions that lack sufficient understanding of family history, underlying tensions, and relational dynamics risk misinterpreting situations, potentially escalating conflict or reinforcing harmful patterns. While AI-mediated interventions may be more appropriate in professional or transactional settings, their application in close family relationships introduces additional risks, where attempts at mediation may inadvertently lead to further harm rather than repair.

Participants in our study expressed diverse and sometimes conflicting views about the appropriateness of chatbot moral judgment. Some participants felt that when harm had occurred, chatbots should clearly articulate moral condemnation—effectively telling a sender, "You should not have said that." Others viewed moral feedback as an opportunity for reflection and increased self-awareness, valuing chatbot interventions that helped them reconsider their own messages. Still others cautioned that overly explicit moral judgment could provoke defensiveness, arguing instead for suggestive or indirect forms of guidance. These divergent perspectives echo our findings on degree of moral authority (Section 4.3.2), underscoring that there is no single acceptable way for chatbots to express moral judgment.

Together, these tensions suggest that designers of chatbot interventions for intergenerational communication must carefully consider not only whether chatbots should make moral judgments, but also how such judgments are conveyed.

## 5.2 Which Normative Behaviours and Expectations Should Be Encoded?

Normative behaviours and expectations are not universal, rather, they are shaped by broader cultural contexts. Prior cross-cultural research suggests that family communication norms are closely tied to culturally specific models of the self. For instance, in many Eastern cultural contexts, intergenerational communication emphasizes relational harmony, filial obligation, and emotional restraint, whereas Western cultural contexts tend to prioritize individual happiness, authenticity, and direct self-expression [33, 42, 67].

These differences have important implications for how chatbot interventions in intergenerational communication are perceived and evaluated. In cultural contexts that emphasize harmony and filial piety, participants may expect chatbots to downplay direct confrontation, foreground benevolent intent, and promote mutual understanding—aligning with roles such as neutral mediators or emotion regulators that soften tension and preserve relational continuity. By contrast, in cultural contexts that value individual autonomy and emotional authenticity, chatbots may be expected to engage more explicitly in boundary setting, to name emotional harm, and to validate recipients' feelings, even if doing so risks short-term relational discomfort.

Our findings suggest that chatbots were often positioned as moral advisors that evaluated communicative appropriateness and guide interaction accordingly. However, which moral principles should be encoded, be it to prioritize relational harmony or individual emotional well-being, for example, cannot be assumed to be culturally neutral. Designers of chatbot interventions for intergenerational communication therefore face a critical design tension: deciding which normative values should guide chatbot behavior, and indeed whether (or how) these values should be made adaptable or configurable across cultural contexts.

## 5.3 Should Chatbots Communicate 'Empathy'?

Participants frequently designed chatbots to communicate empathy, using empathetic language to legitimize emotional experiences and to soften the interpersonal impact of intergenerational communication breakdowns. Participants often described such interventions as "feeling good," suggesting that empathetic responses were perceived as comforting and supportive, particularly when recipients experienced emotional hurt or when senders' emotional motivations required acknowledgment. This aligns with prior findings showing that empathetic chatbot responses can increase users' perceived intimacy with the system, thereby enhancing satisfaction and reuse intention, as demonstrated in mental health counseling contexts [46].

However, prior research also highlights important tensions in deploying empathy within conversational agents. Importantly, chatbots do not possess genuine emotional experiences, and empathetic expressions may therefore be perceived as performative or inauthentic. Recent work by Seitz [53] demonstrates that while empathetic expressions can increase perceived warmth, they may simultaneously decrease perceived authenticity in healthcare chatbots, ultimately undermining user trust. Similarly, Morris et al. [44] show that even when response content is identical, users tend to prefer empathetic messages labeled as coming from a human or peer rather than from an automated system, suggesting that the source of empathy critically shapes how it is interpreted.

These findings point to a design tension that is particularly salient in intergenerational family communication. On the one hand, our participants' designs suggest that empathy plays a meaningful role in emotional repair and relationship maintenance, especially in contexts where family ties are enduring and emotionally charged. On the other hand, overtly human-like expressions of empathy risk being perceived as inauthentic or overstepping the chatbot's legitimate role, potentially eroding trust. An interesting future direction would be to explore whether these messages of empathy from the chatbot are perceived in the way they are intended, or as inauthentic, as in other contexts [44, 53]. Perhaps that value of the empathetic intent can be preserved without resorting to mechanical expressions of empathy.

## 5.4 Should Chatbots Infer Beyond the Conversation?

Participants designed chatbots that, beyond the explicit content of the conversation, made inferences about the conversants' intentions, motivations, or emotional states. These inferences were not always directly supported by the scenario text, yet participants treated them as reasonable grounds for intervention. For example, in S1, when the grandfather criticized his grandson's dislike of bitter gourd by attributing it to generational pickiness, participants often designed chatbots that inferred the grandfather's underlying intention as wanting the grandson to develop healthier eating habits, which is an interpretation not explicitly stated in the scenario. Similar inferential moves appeared in other scenarios, targeting not only senders’ intentions but also recipients’ emotional states, such as assuming that recipients must feel hurt or discouraged and designing interventions to comfort them accordingly.

These designs suggest that participants viewed inference as a way to make chatbot interventions more contextually appropriate and emotionally helpful. By filling in perceived gaps in the conversational record, chatbots could offer explanations, reassurance, or guidance that aligned more closely with participants’ understanding of family dynamics. In this sense, inference functioned as a mechanism for moral and relational sense-making, allowing chatbots to intervene in ways that felt more humane and situationally grounded than responses limited strictly to surface-level message content.

At the same time, inferring beyond what is explicitly stated introduces important risks. Prior work suggests that AI systems that infer users’ intentions, emotions, or personal attributes may evoke feelings of being surveilled or over-analyzed, undermining users’ sense of autonomy and comfort [4, 7]. Moreover, incorrect or overconfident inferences can erode trust, particularly when users feel that the system has misunderstood their situation or motives [50]. In intergenerational family contexts, where emotional stakes are high and relational histories are complex, such misattributions may be especially consequential.

These findings surface a key design tension for chatbot-mediated intergenerational communication: whether interventions should be grounded strictly in the observable conversational content, or whether chatbots should be allowed to make bounded inferences about intent and emotional state. If inference is employed, designers must consider how to do so within reason, such as expressing uncertainty, offering interpretations as possibilities rather than facts, or making the inferential basis transparent, to reduce the likelihood and impact of error. Rather than asking whether chatbots should infer at all, our findings suggest that the central design challenge lies in determining how much, how explicitly, and with what safeguards inference should be incorporated to support helpful intervention without compromising trust.

## 5.5 Limitations / Future Work

This study has several limitations that point to important directions for future research.

First, our scenario-based design interview approach necessarily constrained the scope of inquiry to four predefined scenarios. Although grounded in prior literature on intergenerational accommodation problems, these scenarios do not capture the full range of communication breakdowns that occur in everyday family interactions. To make breakdowns salient, each scenario featured a single sender, whereas real-world tensions often emerge through more distributed or reciprocal exchanges. Future work could explore a broader and more complex set of scenarios, including multi-party and evolving conversational dynamics.

Second, participants designed chatbot interventions within four predefined conditions varying by intervention target and timing. While this structure enabled systematic comparison, it may have limited participants’ exploration of alternative or hybrid forms of AI mediation. Future studies could adopt more open-ended design tasks to surface a wider range of intervention configurations.

Third, our participant sample was shaped by social-media-based recruitment, introducing self-selection bias. Moreover, families experiencing severe or ongoing intergenerational conflict may have been less likely to participate together, meaning that the communication breakdowns explored in this study may reflect relatively moderate tensions rather than deeply strained relationships. Although we recruited across a range of intergenerational relationships (e.g., parent-child, grandparent-grandchild, aunt/uncle-niece/nephew), the final sample consisted entirely of parent-child pairs. In addition, the participant pool exhibited a noticeable gender imbalance, with substantially more female than male participants. Within the scope of this study, we did not observe meaningful differences between younger and elder participants in how they designed chatbot interventions or articulated role expectations, which may reflect shared family norms as well as the sample’s size and cultural homogeneity. Future research could address these limitations by recruiting participants from a wider range of intergenerational relationship types, and aiming for more balanced gender representation. Longitudinal or in-situ deployments of AI-mediated interventions may also help capture how chatbot roles and moral expectations evolve over time within real-world family communication.

All participants were local residents. Although these participants come from a variety of racial and cultural backgrounds, they all reside in a non-Western context. Future work should explore extending our findings to other contexts, as cultural norms around family relationships and communication may vary even more significantly. These explorations will provide additional nuance to interpreting and understanding these tensions.

# 6 Conclusion

This work offers a conceptual and exploratory account of how people envision chatbot interventions for addressing intergenerational communication breakdowns in everyday family chat. Through a scenario-based design study with intergenerational pairs, we showed that participants expect chatbots to employ intervention strategies to mediate understanding, provide emotional support, offer evaluative commentary on problematic messages, and guide interaction through behavioral guidance. The chatbot take on distinct interactional roles—such as neutral mediators, message coaches, repair facilitators, and emotion regulators—depending on intervention target and timing. Across these roles, chatbots were consistently positioned as moral advisors that evaluate communicative appropriateness, interpret intent and impact, and exercise normative judgment within family interactions.

Our findings highlight that designing chatbots for intergenerational communication is not merely a matter of improving message clarity or conversational flow. Instead, designers must grapple with deeper questions about how AI systems should enact moral positioning: when to remain neutral versus when to take a stance, how strongly to assert authority, whether to communicate empathy, and how far to infer beyond what is explicitly said. These design tensions are further shaped by cultural norms and relational expectations, underscoring that there is no universally appropriate chatbot behavior for mediating family communication. By treating chatbots as morally situated actors rather than purely functional tools, designers can better support respectful, sustainable, and culturally sensitive intergenerational communication.

## Acknowledgments

We thank Singapore Ministry of Education and the School of Computer and Information Systems at Singapore Management University for their support.

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# A Scenarios of Integenerational Non-Accommodation

We constructed four conversational scenarios grounded in CAT, each representing a distinct type of non-accommodation, as shown in Figure 4.

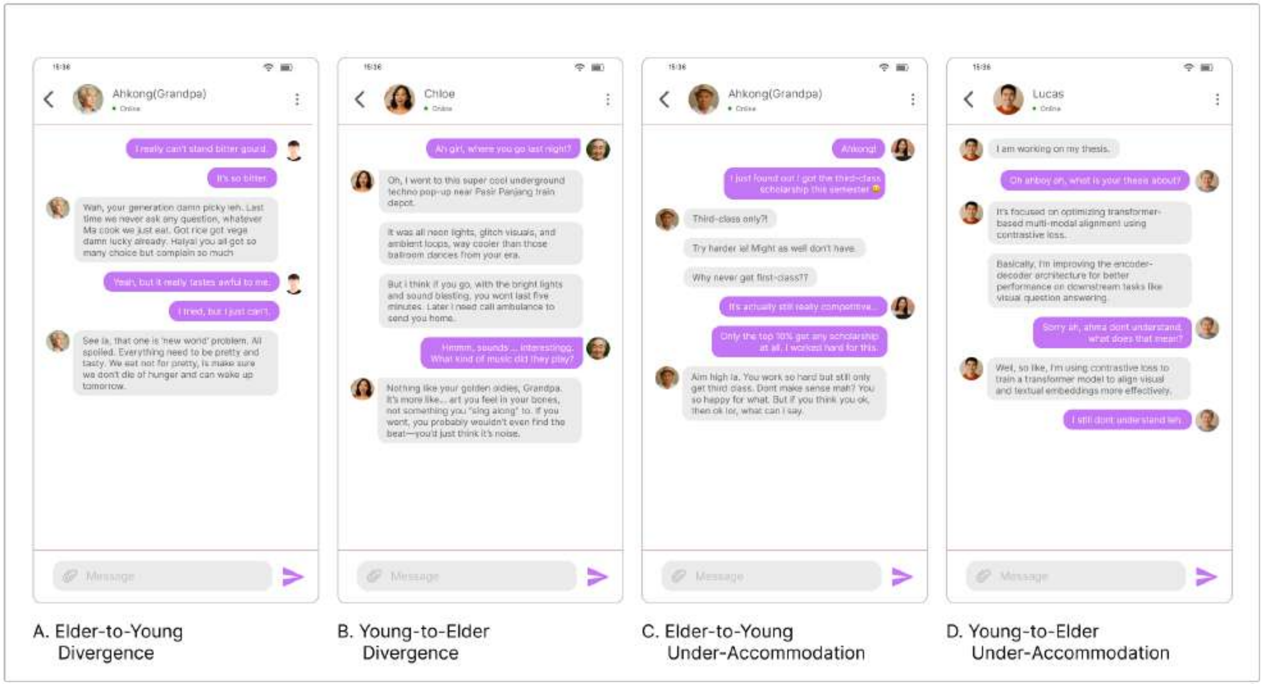


**Figure 4: Four scenarios of intergenerational non-accommodation. (A) The older speaker emphasizes intergenerational distinctions and asserts generational superiority. (B) The younger speaker distances themselves from the older generation by asserting cultural or experiential superiority. (C) The older speaker uses an overly critical and authoritative tone without acknowledging the younger person's emotional investment. (D) The younger speaker uses specialized academic jargon rather than adequately adjusting their explanation to suit the elder's knowledge background.**